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Algebras of Jordan brackets and generalized Poisson algebras

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ABSTRACT

We construct bases for free unital generalized Poisson superalgebras and for free unital superalgebras of Jordan brackets. Also, we prove an analogue of Farkas' theorem for PI unital generalized Poisson algebras and PI unital algebras of Jordan brackets. Relations between generic Poisson superalgebras and superalgebras of Jordan brackets are studied.

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1. Introduction

There are many interesting and important results about the structure of free associative commutative algebras, free associative algebras and free Lie algebras. Although, free Poisson algebras are closely connected with these algebras, only a few results on their structure are available. Free Poisson superalgebras were introduced by Shestakov [1]. For example, the free Poisson algebra of rank 3 and Poisson brackets were used recently in [2,3] to prove that the Nagata automorphism of the polynomial algebra $F[x, y, z]$ over an arbitrary field $\mathbb{F}$ of characteristic 0 is wild. A systematic study of free Poisson algebras was initiated by Makar-Limanov and Umirbaev in 2007 (see [4]) and subsequently continued in [5–8]. A generalization of Poisson algebras known as generic Poisson algebras was studied by in the papers of Kolesnikov et al. [9] and Kaygorodov et al. [10]. Other generalizations of Poisson superalgebras include generalized Poisson superalgebras and superalgebras of Jordan brackets. Every Poisson superalgebra is a superalgebra of Jordan brackets.[11,12] The identities defining unital superalgebras of Jordan brackets were described in [13,14], and an analogue in the non-unital case was obtained in [15]. Superalgebras of Jordan brackets are important in the classification of simple finite-dimensional Jordan superalgebras.[16] Kac and Cantarini [17] studied linearly compact simple superalgebras of Jordan brackets, Kaygorodov and Zhelyabin [18] studied δ -superderivations of simple superalgebras of Jordan brackets, Zelmanov et al. [19] studied the relation between Jordan brackets and Poisson brackets, and special superalgebras of Jordan brackets were considered in [18,20]. Kantor's construction gives interesting relations between Novikov–Poisson algebras and Jordan superalgebras.[21]

In this paper, we construct bases for the free unital generalized Poisson superalgebra with n generators and for the free unital superalgebra of Jordan brackets with n generators. We prove an analogue of Farkas' theorem for PI unital generalized Poisson algebras and PI unital algebras of Jordan brackets. Namely, we prove that every algebra in these varieties satisfying some polynomial identity also satisfies some polynomial of special type. At the end we discuss a relation between generic Poisson superalgebras and superalgebras of Jordan brackets.

2. Definitons and notation

An algebra A over a field $\mathbb{F}$ is called a Jordan algebra whenever it satisfies the identities

$$xy = yx, (x^2y)x = x^2(yx).$$

Denote by G the Grassmann algebra over $\mathbb{F}$ with generators $1, \xi_1, \dots, \xi_n, \dots$ and defining relations $\xi_i^2 = 0$ and $\xi_i \xi_j = -\xi_j \xi_i$. The unity 1 and the products $\xi_{i_1} \cdots \xi_{i_n}$, form a basis for G over $\mathbb{F}$.

Denote by G_0 and G_1 the subspaces generated, respectively, by the products of even and odd length; then G is the direct sum of these subspaces: $G = G_0 \oplus G_1$; moreover, the inclusions $G_i G_j \subseteq G_{i+j \bmod 2}$ hold for $i, j = 0, 1$. In other words, G is a $\mathbb{Z}_2$ -graded algebra (a superalgebra) over $\mathbb{F}$. Now, take an arbitrary superalgebra $A = A_0 \oplus A_1$ over $\mathbb{F}$ and consider the tensor product $G \otimes A$ of $\mathbb{F}$ -algebras. Its subalgebra

$$G(A) = G_0 \otimes A_0 + G_1 \otimes A_1$$

is called the Grassmann envelope of the superalgebra A .

Given a variety Ω of algebras over $\mathbb{F}$, a superalgebra $A = A_0 \oplus A_1$ is called an Ω -superalgebra whenever its Grassmann envelope $G(A)$ is an algebra in Ω .

Recall the notion of the **Kantor double** [12]. Take an associative supercommutative superalgebra $A = A_0 \oplus A_1$ equipped with a super-anticommutative multiplication $\{, \}$: $A \times A \rightarrow A$, which we call the bracket. From A with $\{, \}$ we can construct a superalgebra $K(A)$. Consider the direct sum $K(A) = A \oplus Ax$ of linear spaces, where Ax is an isomorphic copy of the space A . Define a multiplication $*$ on $K(A, \{, \})$ by the relations

$$a * b = ab, \quad a * bx = (ab)x,$$

$$ax * b = (-1)^{|b|}(ab)x, \quad ax * bx = (-1)^{|b|}\{a, b\}$$

for all homogeneous elements a and b of A . Put $K(A)_0 = A_0 \oplus A_1 x$ and $K(A)_1 = A_1 \oplus A_0 x$; therefore, $K(A)$ is a $\mathbb{Z}_2$ -graded algebra.

A superalgebra JB is called a superalgebra of Jordan brackets whenever its Kantor double is a Jordan superalgebra. As [13,14] implies, a superalgebra JB whose associative-supercommutative multiplication $\cdot$ and superanticommutative multiplication $\{, \}$ satisfy

$$\{a, bc\} = \{a, b\}c + (-1)^{|a||b|}b\{a, c\} - D(a)bc, \quad (1)$$

$$\begin{aligned} \{a, \{b, c\}\} &= \{\{a, b\}, c\} + (-1)^{|a||b|}\{b, \{a, c\}\} \\ &\quad + D(a)\{b, c\} + (-1)^{|a||bc|}D(b)\{c, a\} + (-1)^{|c||ab|}D(c)\{a, b\}, \end{aligned} \quad (2)$$

with $D(a) = \{a, 1\}$, is a superalgebra of Jordan brackets. Observe that D is an even derivation of the superalgebra $(JB, \cdot, \{, \})$. If $D = 0$ then $(JB, \cdot, \{, \})$ is a Poisson superalgebra [11]. As [17] implies, on the same vector space we can consider the new multiplication $\{a, b\}_D = \{a, b\} + \frac{1}{2}(aD(b) - D(a)b)$ and the new linear mapping $D_D = \frac{1}{2}D$.

It is a new superalgebra with the identities

$$\{a, bc\}_D = \{a, b\}_D c + (-1)^{|a||b|} b \{a, c\}_D - D_D(a)bc, \quad (3)$$

$$\{a, \{b, c\}_D\}_D = \{\{a, b\}_D, c\}_D + (-1)^{|a||b|} \{b, \{a, c\}_D\}_D. \quad (4)$$

Every superalgebra satisfying (3) and (4) is called a generalized Poisson superalgebra.

All vector spaces are considered over a field $\mathbb{F}$ of characteristic zero.

3. A basis for the free generalized poisson superalgebra

To describe bases for free algebraic structures is one of the main problems in modern algebra. Researchers have constructed bases for free Lie algebras, [22] free Lie superalgebras, [23] partially commutative Lie algebras, [24] free Poisson superalgebras, [1] free generic Poisson algebras, [10] and so on.

Let us now consider free unital generalized Poisson superalgebras. Take the free non-associative superdialgebra $(D_X, \cdot, \{, \})$ over a field $\mathbb{F}$ with the set $X = X_0 \cup X_1 = \{1, x_1, \dots, x_n\}$ of even and odd generators. Consider the ideal I of D_X generated by

$$\left\{ \begin{aligned} &\{a, b\} + (-1)^{|a||b|} \{b, a\}, \quad a \cdot b - (-1)^{|a||b|} b \cdot a, \quad 1 \cdot a - a, \\ &\{a, b\} \cdot c + (-1)^{|a||b|} b \cdot \{a, c\} - D(a) \cdot b \cdot c - \{a, b \cdot c\}, \\ &\{\{a, b\}, c\} + (-1)^{|a||b|} \{b, \{a, c\}\} - \{a, \{b, c\}\} \mid a, b, c \in D_{X_0} \cup D_{X_1} \end{aligned} \right\},$$

with $D(a) = \{a, 1\}$. It is easy to see that D_X/I is a free unital generalized Poisson superalgebra.

Denote by $GenP$ the free unital generalized Poisson superalgebra generated by X , assumed to be a linearly ordered set with $1 < x_1 < \dots < x_n$. Define the set of good words as follows. All elements of X are good words. On assuming that all good words of length less than n are defined, a word w of length n in the alphabet X is a good word whenever the following conditions hold:

- (1) $w = \{u, v\}$, where u and v are good words with $u > v$;
- (2) if $w = \{\{u_1, u_2\}, v\}$ then $u_2 \leq v$.

Define an order on the good words of length less than or equal to n to be consistent with the existing order on words and satisfy the following condition: if w is a good word and $w = \{u, v\}$ then $w > u$ and $u > v$.

Define the set

$$M = \{u, \{v, v\} \mid u, v \text{ are good words in the alphabet } X \text{ and } v \in GenP_1\}$$

and then a linear order on it. Put $u > v$ when $\deg(u) > \deg(v)$; if $\deg(u) = \deg(v)$ then for $u = \{u_1, u_2\}$ and $v = \{v_1, v_2\}$, where $u_1 > u_2$ and $v_1 > v_2$, put $u > v$ when $u_1 > v_1$, and if $u_1 = v_1$ put $u > v$ when $u_2 > v_2$. Denote the elements of M by e_i , where $e_i < e_j$ for $i < j$.

Denote by L_X the free non-associative superalgebra over $\mathbb{F}$ with a set of generators X and multiplication $\{, \}_L$. Consider the ideal J of L_X generated by the set

$$\{ \{a, b\} + (-1)^{|a||b|} \{b, a\}, \{ \{a, b\}, c\} + (-1)^{|a||b|} \{b, \{a, c\}\} - \{a, \{b, c\}\} \mid a, b, c \in L_{X_0} \cup L_{X_1} \}.$$

The quotient L_X/J is a free Lie superalgebra [23]; furthermore,

the set M is a basis for L_X/J .

Denote by K the ideal in D_X generated by the elements of type $\{x \cdot y \mid x, y \neq 1\}$. It is easy to see that $(D_X/I)/K \cong L_X/J$.

Put

$$U = \left\{ e_{i_1}^{k_1} \dots e_{i_n}^{k_n} \mid e_{i_1} \neq 1, e_{i_m} \in M, i_1 < \dots < i_n \right\} \cup \{1\}.$$

This set is closed under the multiplication $\cdot$, which is supercommutative, and so we can consider the elements of M as $e_{i_1}^{k_1} \dots e_{i_n}^{k_n}$, where $k_i \neq 0$, and if $e_{i_j} \in \text{Gen}P_1$ then $k_j = 1$. Say that an element $e_{i_1}^{k_1} \dots e_{i_n}^{k_n}$ is of degree $\deg_e = \sum k_i$. For $e_i \in M$ define $\deg_L$ as the length of the word e_i in the generators X . The degree of an element f is the number of generators in the word for f .

The main goal of this chapter is to find a basis for the free unital generalized Poisson superalgebra generated by X . We show that U is a basis for the free unital generalized Poisson superalgebra generated by $\{1, x_1, \dots, x_n\}$.

Lemma 1: *The set U is a basis for the free superalgebra $\text{Gen}P$ as a vector space.*

Proof: Consider a degree 2 word w . It is easy to see that w is of type $x_i x_j$ or type $\{x_i, x_j\}$ or type $\{1, x_i\}$, and $w \in U$. We claim that every word of degree n is a linear combination of the elements of U . Hence, every element of the superalgebra $\text{Gen}P$ lies in the linear span of U . We justify this by induction on n . Assume that every word of degree less than n lies in the linear span of U . Then $w = w_1 w_2$ or $w = \{w_1, w_2\}$, where w_1 and w_2 are words of degrees less than n , that is, w_1 and w_2 lie in the linear span of U . Consider two cases.

The first case: $w_1 = \sum_i \alpha_{1i} m_{1i}$ and $w_2 = \sum_i \alpha_{2i} m_{2i}$, where $m_{1i}, m_{2i} \in U, \alpha_{1i}, \alpha_{2i} \in k$. Hence, $w = \sum_{i,j} \alpha_{1i} \alpha_{2j} m_{1i} m_{2j}$, and, since U is closed under the multiplication $\cdot$, we see that $w = \sum_i \alpha_i m_i, m_i \in U, \alpha_i \in k$.

The second case: $w_1 = \sum_i \alpha_{1i} m_{1i}$ and $w_2 = \sum_i \alpha_{2i} m_{2i}$, where $m_{1i}, m_{2i} \in U, \alpha_{1i}, \alpha_{2i} \in k$. Therefore, $w = \sum_{i,j} \alpha_i \alpha_j \{m_i, m_j\}$. Consider an arbitrary term $\{m_i, m_j\}$. If m_i or m_j is equal to $n_1 n_2$, where $n_i \in U$ with $n_i \neq 1$, then m_j is of type $n_1 n_2$. Use the relation

$$\{m_i, n_1 n_2\} = \{m_i, n_1\} n_2 + (-1)^{|m_i||n_1|} n_1 \{m_i, n_2\} - D(m_i) n_1 n_2.$$

Here we observe that the degrees of $\{m_i, n_1\}$, $\{m_i, n_2\}$, and $D(m_i)$ are less than n and $n_1, n_2 \in U$. It is easy to see that $\{m_i, n_1\}$, $\{m_i, n_2\}$, and $D(m_i)$ lie in the linear span of U ; hence, so does w .

If neither m_i nor m_j is of type $n_1 n_2$, where $n_1, n_2 \in U$ and $n_1, n_2 \neq 1$, then $m_i, m_j \in M$. If $m_i = m_j \in \text{Gen}P_1$ then $\{m_i, m_j\} \in M$. If $m_i \neq m_j$ then we can say that $m_i > m_j$, and if $m_i = \{m_{1i}, m_{2i}\}$ with $m_{1i} \geq m_{2i}$ then for $m_{2i} \leq m_j$ we have $\{m_i, m_j\} \in M$. For $m_{2i} > m_j$ we have

$$\{m_i, m_j\} = \{\{m_{1i}, m_j\}, m_{2i}\} + (-1)^{|m_{1i}||m_j|} \{m_{1i}, \{m_{2i}, m_j\}\},$$

where $\{\{m_{1i}, m_j\}, m_{2i}\} \in M$. If $\{m_{1i}, \{m_{2i}, m_j\}\} \in M$ then we infer that $\{m_i, m_j\}$ lies in the linear span of U . Otherwise, we can consider m_{1i} as the $\{\cdot, \cdot\}$ -product of m_{11i} and m_{21i} . Following that, we can continue the process using the identity (4). Observe that the degree of m_i is greater than that of m_{1i} , which is greater than that of m_{11i} , and so on. Therefore, the process is finite. $\square$

Given $b = \prod_{k=1}^n e_k^{t_k}$, put $\frac{b}{e_m} = e_1^{t_1} \cdots e_{m-1}^{t_{m-1}} e_m^{t_m-1} e_{m+1}^{t_{m+1}} \cdots e_n^{t_n}$.

It is easy to see that

$$\begin{aligned} \{a, e_k^{t_k}\} &= \{a, e_k^{t_k-1}\} e_k + (-1)^{|a||e_k^{t_k-1}|} e_k^{t_k-1} \{a, e_k\} - D(a) e_k^{t_k} \\ &= \{a, e_k^{t_k-2}\} e_k^2 + (-1)^{|a||e_k^{t_k-2}|} e_k^{t_k-2} \{a, e_k\} e_k + (-1)^{|a||e_k^{t_k-1}|} e_k^{t_k-1} \{a, e_k\} - 2D(a) e_k^{t_k} \\ &\quad \dots \\ &= t_k \{a, e_k\} e_k^{t_k-1} - (t_k - 1) D(a) e_k^{t_k}. \end{aligned}$$

Using this notation, we obtain

$$\begin{aligned} \{a, b\} &= \{a, \prod_{k=1}^n e_k^{t_k}\} = \left\{a, \frac{b}{e_n^{t_n}}\right\} e_n^{t_n} + (-1)^{|a||\frac{b}{e_n^{t_n}}|} \frac{b}{e_n^{t_n}} \{a, e_n^{t_n}\} - D(a) b \\ &= \left\{a, \frac{b}{e_{n-1}^{t_{n-1}} e_n^{t_n}}\right\} e_{n-1}^{t_{n-1}} e_n^{t_n} + (-1)^{|a||\frac{b}{e_{n-1}^{t_{n-1}} e_n^{t_n}}|} \frac{b}{e_{n-1}^{t_{n-1}} e_n^{t_n}} \{a, e_{n-1}^{t_{n-1}}\} e_n^{t_n} \\ &\quad + (-1)^{|a||\frac{b}{e_n^{t_n}}|} \frac{b}{e_n^{t_n}} \{a, e_n^{t_n}\} - 2D(a) b \\ &= \dots \\ &= \sum_{k=1}^n (-1)^{|e_1^{t_1} \cdots e_{k-1}^{t_{k-1}}| |e_k^{t_k}|} t_k \{a, e_k\} \frac{b}{e_k} - \left(\sum_{k=1}^n t_k - 1\right) D(a) b. \end{aligned}$$

Define a linear mapping $*$: $U \rightarrow U$ as follows:

- (1) if $w \in X$ then $w^* = w$;
- (2) if $w \in M$ then $w^* = w$;
- (3) if $a = \prod_{k=1}^n e_k^{a_k}, b = \prod_{k=1}^n e_k^{b_k}$ then

$$(ab)^* = (-1)^{\sum_{i=1}^n |e_i^{a_i}| |e_{i+1}^{b_{i+1}} \cdots e_n^{b_n}|} \prod_{k=1}^n e_k^{a_k + b_k}.$$

- (4) Assume that we have defined $*$ for all words of length less than n . Consider the word $w = \{w_1, w_2\}$ with $w_1, w_2 \in M$ as an element of the free Lie superalgebra $(M, \{\cdot, \cdot\})$. Define w^* as a linear combination of the basis elements of $(M, \{\cdot, \cdot\})$, that is, w^* lies in the linear span of M .

- (5) Assume that we have defined $*$ for all words of length less than m . Without loss of generality, for the word $w = \{w_1, w_2\}$ of length m , where w_1 or w_2 lies in $U \setminus M$, we have $w_2 \in U \setminus M$ and $w_2 = \Pi_{k=1}^n e_k^{t_k}$. Then

$$(\{w_1, w_2\})^* = \sum_{k=1}^n (-1)^{|e_1^{t_1} \cdots e_{k-1}^{t_{k-1}}| |e_k^{t_k}|} t_k \left(\{w_1, e_k\}^* \frac{w_2}{e_k} \right)^* - \left(\sum_{k=1}^n t_k - 1 \right) (D(w_1)^* w_2)^*,$$

where the elements of type $\{x, y\}^*$ in the right-hand side are defined above.

Theorem 2: *The set U is a basis for the free superalgebra $\text{Gen}P$.*

Proof: Consider U as a vector space basis and define two new multiplications $*$ and $\{, \}^*$ as $u * v = (uv)^*$ and $\{u, v\}^* = (\{u, v\})^*$. This yields the new superalgebra $(U, *, \{, \}^*)$. We claim that it is a generalized Poisson superalgebra. Then we infer that U is a basis for $\text{Gen}P$.

It is easy to see that $*$ is associative and supercommutative, while $\{, \}^*$ is superanticommutative. We claim that $(U, *, \{, \}^*)$ satisfies the identity

$$\{a, b * c\}^* = \{a, b\}^* * c + (-1)^{|a||b|} b * \{a, c\}^* - D(a)^* * b * c, \quad (5)$$

and establish it by induction on $\deg_e(a) + \deg_e(b) + \deg_e(c)$.

Observe that

$$\begin{aligned} \{1, e_i * e_j\}^* &= \{1, e_i\}^* * e_j + e_i * \{1, e_j\}^*, \\ \{e_k, e_i * e_j\}^* &= \{e_k, e_i\}^* * e_j + (-1)^{|e_k||e_i|} e_i * \{e_k, e_j\}^* - D(e_k)^* * e_i * e_j \end{aligned}$$

by the definition of the mapping $*$. Hence, if $\deg_e(a) + \deg_e(b) + \deg_e(c) < 4$ then (5) holds.

Suppose that $\deg_e(a) + \deg_e(b) + \deg_e(c) > 3$. For $b = \Pi_{k=1}^n e_k^{b_k}$ and $c = \Pi_{k=1}^n e_k^{c_k}$ put

$$\gamma(b, c) = \sum_{i=1}^n |e_i^{b_i}| |e_{i+1}^{c_{i+1}} \cdots e_n^{c_n}| \text{ and } \mu(b, c, k) = |e_1^{b_1+c_1} \cdots e_{k-1}^{b_{k-1}+c_{k-1}}| |e_k^{b_k+c_k}|.$$

Hence,

$$D(a) * b * c = (-1)^{\gamma(b, c)} D(a) * \Pi_{k=1}^n e_k^{b_k+c_k}. \quad (6)$$

We can see that

$$\begin{aligned} (-1)^{\gamma(b, c)} \{a, b * c\}^* &= \sum_{k=1}^n (-1)^{\mu(b, c, k)} (b_k + c_k) \{a, e_k\}^* * \frac{\Pi_{m=1}^n e_m^{b_m+c_m}}{e_k} \\ &\quad - \left(\sum_{k=1}^n (b_k + c_k) - 1 \right) D(a)^* * \Pi_{m=1}^n e_m^{b_m+c_m}. \end{aligned} \quad (7)$$

Furthermore,

$$\begin{aligned}
 & \{a, b\}^* * c \\
 &= \sum_{k=1}^n (-1)^{|e_1^{b_1} \cdots e_{k-1}^{b_{k-1}}| |e_k^{b_k}|} b_k \{a, e_k\}^* * \frac{b}{e_k} * c - \left(\sum_{k=1}^n b_k - 1 \right) D(a)^* * b * c \\
 &= \sum_{k=1}^n (-1)^{|e_1^{b_1} \cdots e_{k-1}^{b_{k-1}}| |e_k^{b_k}| + \sum_{i=1}^n |e_i^{b_i}| |e_{i+1}^{c_{i+1}}| \cdots |e_n^{c_n}| - |e_k| |e_1^{c_1} \cdots e_{k-1}^{c_{k-1}}|} b_k \{a, e_k\}^* * \frac{\prod_{m=1}^n e_m^{b_m+c_m}}{e_k} \\
 &\quad - (-1)^{\sum_{i=1}^n |e_i^{b_i}| |e_{i+1}^{c_{i+1}}| \cdots |e_n^{c_n}|} \left(\sum_{k=1}^n b_k - 1 \right) D(a)^* * \prod_{m=1}^n e_m^{b_m+c_m}.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \{a, b\}^* * c &= (-1)^{\gamma(b,c)} \left(\sum_{k=1}^n (-1)^{\mu(b,c,k)} b_k \{a, e_k\}^* * \frac{\prod_{m=1}^n e_m^{b_m+c_m}}{e_k} \right. \\
 &\quad \left. - \left(\sum_{k=1}^n b_k - 1 \right) D(a)^* * \prod_{m=1}^n e_m^{b_m+c_m} \right). \quad (8)
 \end{aligned}$$

Observe that

$$\begin{aligned}
 & (-1)^{|c||b|} \{a, c\}^* * b \\
 &= (-1)^{|c||b|} \sum_{k=1}^n (-1)^{|e_1^{c_1} \cdots e_{k-1}^{c_{k-1}}| |e_k^{c_k}|} c_k \{a, e_k\}^* * \frac{c}{e_k} * b - \left(\sum_{k=1}^n c_k - 1 \right) D(a)^* * b * c \\
 &= \sum_{k=1}^n (-1)^{|e_k^{c_k}| |b| + |e_k| |e_1^{c_1} \cdots e_{k-1}^{c_{k-1}}| + \sum_{i=1}^n |e_i^{b_i}| |e_{i+1}^{c_{i+1}}| \cdots |e_n^{c_n}| - |e_k| |e_{k+1}^{b_{k+1}} \cdots e_n^{b_n}|} b_k \{a, e_k\}^* * \frac{\prod_{m=1}^n e_m^{b_m+c_m}}{e_k} \\
 &\quad - (-1)^{\sum_{i=1}^n |e_i^{b_i}| |e_{i+1}^{c_{i+1}}| \cdots |e_n^{c_n}|} \left(\sum_{k=1}^n c_k - 1 \right) D(a)^* * \prod_{m=1}^n e_m^{b_m+c_m}.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 (-1)^{|c||b|} \{a, c\}^* * b &= (-1)^{\gamma(b,c)} \left(\sum_{k=1}^n (-1)^{\mu(b,c,k)} c_k \{a, e_k\}^* * \frac{\prod_{k=1}^n e_m^{b_m+c_m}}{e_k} \right. \\
 &\quad \left. - \left(\sum_{k=1}^n c_k - 1 \right) D(a)^* * \prod_{m=1}^n e_m^{b_m+c_m} \right). \quad (9)
 \end{aligned}$$

Finally, adding (6) and (7), then subtracting (8) and (9), we see that $(U, *, \{, \}^*)$ satisfies (5).

Now we claim that this superalgebra satisfies the identity

$$\{a, \{b, c\}^*\}^* = \{\{a, b\}^*, c\}^* + (-1)^{|a||b|} \{b, \{a, c\}^*\}^*. \quad (10)$$

For $a, b, c \in M$ this holds because $(M, \{, \}^*)$ is a Lie superalgebra.

Without loss of generality, take $a = a_1 * a_2$ with $a_1, a_2 \in U$ and $a_1, a_2 \neq 1$. Using (5), we obtain

$$\begin{aligned}
 & \{\{a_1 * a_2, b\}^*, c\}^* + (-1)^{|a||b|} \{b, \{a_1 * a_2, c\}^*\}^* \\
 &= -(-1)^{|b||a|} \{\{b, a_1\}^* * a_2 + (-1)^{|b||a_1|} a_1 * \{b, a_2\}^* - D(b)^* a, c\}^* \\
 &\quad - (-1)^{|a|(|b|+|c|)} \{b, \{c, a_1\}^* * a_2 + (-1)^{|c||a_1|} a_1 * \{c, a_2\}^* - D(c)^* * a_1 * a_2\}^* \\
 &= (-1)^{|b||a|+|c|(|a|+|b|)} (\{c, \{b, a_1\}^*\}^* * a_2 + (-1)^{(|b|+|a_1|)|c|} \{b, a_1\}^* * \{c, a_2\}^* \\
 &\quad - D(c)^* * \{b, a_1\}^* * a_2) \\
 &\quad + (-1)^{|b||a|+|c|(|b|+|a|)+|b||a_1|} (\{c, a_1\}^* * \{b, a_2\}^* + (-1)^{|c||a_1|} a_1 * \{c, \{b, a_2\}^*\}^* \\
 &\quad - D(c)^* * a_1 * \{b, a_2\}^*) \\
 &\quad + (-1)^{|b||a|+|c|(|a|+|b|)} (\{c, D(b)^*\}^* * a + (-1)^{|c||b|} D(b)^* * \{c, a\}^* \\
 &\quad - D(c)^* * D(b)^* * a) \\
 &\quad - (-1)^{|a|(|b|+|c|)} (\{b, \{c, a_1\}^*\}^* * a_2 + (-1)^{|b|(|c|+|a_1|)} \{c, a_1\}^* * \{b, a_2\}^* \\
 &\quad - D(b)^* * \{c, a_1\}^* * a_2) \\
 &\quad - (-1)^{|a|(|b|+|c|)+|c||a|} (\{b, a_1\}^* * \{c, a_2\}^* + (-1)^{|b||a_1|} a_1 * \{b, \{c, a_2\}^*\}^* \\
 &\quad - D(b)^* * a_1 * \{c, a_2\}^*) \\
 &\quad - (-1)^{|a|(|b|+|c|)} (\{b, D(c)^*\}^* * a + (-1)^{|b||c|} D(c)^* * \{b, a\}^* - D(b)^* * D(c)^* * a) \\
 &= -(-1)^{|a|(|b|+|c|)} (\{\{b, a_1\}^*, c\}^* + (-1)^{|b||a_1|} \{b, \{c, a_1\}^*\}^*) * a_2 \\
 &\quad + (-1)^{(|b|+|c|)|a_1|} a_1 * (\{\{b, a_2\}^*, c\}^* + (-1)^{|b||a_2|} \{b, \{c, a_2\}^*\}^*) - \{D(b)^*, c\}^* * a \\
 &\quad - \{b, D(c)^*\}^* * a \\
 &= -(-1)^{|a||bc|} (\{\{b, c\}^*, a_1\}^* * a_2 + (-1)^{|bc||a_1|} a_1 * \{\{b, c\}^*, a_2\}^* \\
 &\quad - D(\{b, c\}^*)^* * a = \{a_1 * a_2, \{b, c\}^*\}^*.
 \end{aligned}$$

Therefore, the superalgebra satisfies (10), and the proof of the theorem is complete. $\square$

Denote by $GenP(n)$ the multi-homogeneous component of the free generalized Poisson algebra $GenP$ on $n + 1$ generators $1, x_1, \dots, x_n$.

Theorem 3: We have $\dim(GenP(n)) = n \cdot n!$.

Proof: Consider $GenP[1, x_1, \dots, x_n]$ as a linear subspace of the free Poisson algebra $P[1, x_1, \dots, x_n]$ on $n + 1$ generators. It is known that the dimension of the multi-homogeneous component of the free Poisson algebra with $n + 1$ generators is $(n + 1)!$. Using the description of a basis for the free generalized Poisson algebra (Theorem 2), we can find a basis for the multi-homogeneous component. Therefore, a basis for the multi-homogeneous component of the free generalized Poisson algebra with generators $1, x_1, \dots, x_n$ and a basis for the multi-homogeneous component of the free Poisson algebra with the same generators differ only in the elements of type $1 \cdot y$, where y is an element of the basis for the free Poisson algebra with generators $x_1, \dots, x_n$. Thus, $\dim(GenP(n)) = (n + 1)! - n! = n \cdot n!$. $\square$

4. A basis for the free superalgebra of Jordan brackets

Consider free superalgebras of Jordan brackets. Denote by $(D_X, \cdot, \{, \})$ the free non-associative super-dialgebra over $\mathbb{F}$ with the set of even and odd generators $X = X_0 \cup X_1 = \{1, x_1, \dots, x_n\}$.

Consider the ideal I of D_X generated by the set

$$\begin{aligned} & \{ \{a, b\} + (-1)^{|a||b|} \{b, a\}, a \cdot b - (-1)^{|a||b|} b \cdot a, 1 \cdot a - a, \\ & \{a, b\} \cdot c + (-1)^{|a||b|} b \cdot \{a, c\} - D(a) \cdot b \cdot c - \{a, b \cdot c\}, \\ & \{ \{a, b\}, c \} + (-1)^{|a||b|} \{b, \{a, c\}\} - \{a, \{b, c\}\} - D(a)\{b, c\} - (-1)^{|a||bc|} D(b)\{c, a\} \\ & \quad - (-1)^{|c||ab|} D(c)\{a, b\} | \\ & a, b, c \in D_{X_0} \cup D_{X_1} \}, \end{aligned}$$

with $D(a) = \{a, 1\}$. It is easy to see that D_X/I is a free unital superalgebra of Jordan brackets.

Denote by JB the free unital superalgebra of Jordan brackets generated by X , assumed to be a linearly ordered set with $1 < x_1 < \dots < x_n$.

Define the set of good words as follows. All elements of X are good words. Assume that we have defined all good words of length less than n . A word w of length n in the alphabet X is a good word whenever:

- (1) $w = \{u, v\}$, where u and v are good words and $u > v$;
- (2) $w = \{\{u_1, u_2\}, v\}$ with $u_2 \leq v$.

Define an order on good words of length less than or equal to n to be consistent with the existing order on words and satisfy the following condition: if w is a good word and $w = \{u, v\}$ then $w > u, v$.

Put

$$M = \{u, \{v, v\} | u, v \text{ are good words in the alphabet } X, v \in JB_1\}.$$

To define a linear order on this set, put $u > v$ whenever $\deg(u) > \deg(v)$; if $\deg(u) = \deg(v)$ then for $u = \{u_1, u_2\}$ and $v = \{v_1, v_2\}$ with $u_1 > u_2$ and $v_1 > v_2$ put $u > v$ whenever $u_1 > v_1$, and if $u_1 = v_1$ then put $u > v$ whenever $u_2 > v_2$. Denote the elements of M by e_i , where $e_i < e_j$ for $i < j$.

Denote by L_X the free non-associative superalgebra over $\mathbb{F}$ with a set of generators X and multiplication $\{, \}_L$. Consider the ideal J of L_X generated by the set

$$\{ \{a, b\} + (-1)^{|a||b|} \{b, a\}, \{ \{a, b\}, c \} + (-1)^{|a||b|} \{b, \{a, c\}\} - \{a, \{b, c\}\} | a, b, c \in L_{X_0} \cup L_{X_1} \}.$$

The quotient L_X/J is a free Lie superalgebra [23]; furthermore, the set M is a basis for it.

Denote by K the ideal of D_X generated by the elements of type $\{x \cdot y | x, y \neq 1\}$. It is easy to see that $(D_X/I)/K \cong L_X/J$.

Put

$$U = \{e_{i_1}^{k_1} \dots e_{i_n}^{k_n} | e_{i_1} \neq 1, e_{i_m} \in M, i_1 < \dots < i_n\} \cup \{1\}.$$

This set is closed under the multiplication $\cdot$, which is supercommutative, and so we can regard the elements of M as $e_{i_1}^{k_1} \dots e_{i_n}^{k_n}$ with $k_i \neq 0$, and if $e_{i_j} \in JB_1$ then $k_j = 1$. Define the

degree of an element $e_{i_1}^{k_1} \dots e_{i_n}^{k_n}$ as $\deg_e = \sum k_i$. Given $e_i \in M$, define $\deg_L$ as the length of this word in the generators X .

The main goal of this chapter is to describe a basis for the free superalgebra of Jordan brackets. We show that U is a basis for the free superalgebra of Jordan brackets.

Lemma 4: *The set U is a basis for the free superalgebra JB as a vector space.*

Proof: It is similar to the proof of Lemma 1.

Define the linear mapping $*$: $U \rightarrow U$ as follows:

- (1) if $w \in X$ then $w^* = w$;
- (2) if $w \in M$ then $w^* = w$;
- (3) if $a = \prod_{k=1}^n e_k^{a_k}$, $b = \prod_{k=1}^n e_k^{b_k}$ then

$$(ab)^* = (-1)^{\sum_{i=1}^n |e_i^{a_i}| |e_{i+1}^{b_{i+1}}| \dots |e_n^{b_n}|} \prod_{k=1}^n e_k^{a_k+b_k}.$$

- (4) Assume that we have defined $*$ on all words of length less than n and all elements u with $u < w$. Consider the word $w = \{w_1, w_2\}$ with $w_1 = \{w_{11}, w_{12}\}$, where $w_1, w_2, w_{11}, w_{12} \in M$ and $\{w_{11}, w_{12}\} = w_1$ with $w_{11}, w_{12} > w_2$, as an element of the superalgebra $(M, \{, \})$. Using (2), define w^* as a linear combination of the elements of

$$W = \{\{w_{11}, w_2\}, w_{12}\}, \{w_{12}, w_2\}, w_{11}\}, D(w_2) \cdot \{w_{11}, w_{12}\}, D(w_{11}) \cdot \{w_2, w_{12}\}, D(w_{12}) \cdot \{w_2, w_{11}\}\},$$

where either $u \in W$ and $u < w$ or $u \in W$ and $u = u_1 \cdot u_2$, where $u_1, u_2 < w$. This means that for every $u \in W$ we define u^* .

- (5) Assume that we have defined $*$ on all words of length less than m . For the word $w = \{w_1, w_2\}$ of length m , where w_1 or w_2 is an element of $U \setminus M$, we can think that $w_2 \in U \setminus M$ and $w_2 = \prod_{k=1}^n e_k^{t_k}$. Then

$$(\{w_1, w_2\})^* = \sum_{k=1}^n (-1)^{|e_1^{t_1}| \dots |e_{k-1}^{t_{k-1}}| |e_k^{t_k}|} t_k \left(\{w_1, e_k\}^* \frac{w_2}{e_k} \right)^* - \left(\sum_{k=1}^n t_k - 1 \right) (D(w_1)^* w_2)^*,$$

where the elements of type $\{x, y\}^*$ in the right-hand side are defined above.

□

Theorem 5: *The set U is a basis for the free superalgebra JB .*

Proof: It is similar to the proof of Theorem 2.

Denote by $JB(n)$ the multi-homogeneous component of the free unital algebra of Jordan brackets JB on $n + 1$ generators $1, x_1, \dots, x_n$. □

Theorem 6: *We have $\dim(JB(n)) = n \cdot n!$.*

Proof: This is similar to the proof of Theorem 3.

Combining Theorems 2 and 5 with [17], we obtain the following assertion. □

Theorem 7: *A unital superalgebra of Jordan brackets $(A, \cdot, \{, \})$ is free (or simple) if and only if so is the unital generalized Poisson superalgebra $(A, \cdot, \{, \}_D)$.*

5. Analogues of Farkas' theorem for PI algebras

The celebrated Amitsur–Levitsky theorem states that the algebra $M_k(R)$ of $k \times k$ matrices over a commutative ring R satisfies the identity $s_{2k} = 0$. An analogue of the Amitsur–Levitsky theorem for matrix superalgebras was established in [25]. Furthermore, every associative PI algebra satisfies $(s_k)^l = 0$ by Amitsur's theorem. Poisson PI algebras were studied in [26–30]. D. Farkas defined customary polynomials

$$g = \sum_{\sigma \in S_m} c_\sigma \{x_{\sigma(1)}, x_{\sigma(2)}\} \dots \{x_{\sigma(2i-1)}, x_{\sigma(2i)}\}$$

and proved that every Poisson PI algebra satisfies some customary identity [26]. Farkas' theorem was established for generic Poisson algebras in [9]. There is an analogue of customary identities for generalized Poisson algebras and algebras of Jordan brackets:

$$g_* = \sum_{i=0}^{[m/2]} \sum_{\sigma \in S_m} c_{\sigma,i} \langle x_{\sigma(1)}, x_{\sigma(2)} \rangle \dots \langle x_{\sigma(2i-1)}, x_{\sigma(2i)} \rangle D(x_{\sigma(2i+1)}) \dots D(x_{\sigma(m)}), \quad (11)$$

where

$$\langle x, y \rangle := \{x, y\} - (D(x)y - xD(y)).$$

It is easy to see that the algebra $(JB, \cdot, \langle, \rangle)$ satisfies the identities

$$\begin{aligned} \langle x, y \rangle &= -\langle y, x \rangle, \\ \langle x, y \cdot z \rangle &= \langle x, y \rangle \cdot z + y \cdot \langle x, z \rangle, \\ \langle \langle x, y \rangle, z \rangle + \langle \langle y, z \rangle, x \rangle + \langle \langle z, x \rangle, y \rangle &= D(x)\{y, z\} + D(y)\{z, x\} + D(z)\{x, y\}. \end{aligned}$$

Therefore, $(JB, \cdot, \langle, \rangle)$ is a generic Poisson algebra (see [10]). It is shown in [20] that if the Kantor double of a superalgebra $(JB, \cdot, \{, \})$ is a special Jordan superalgebra then $(JB, \cdot, \langle, \rangle)$ satisfies the relation

$$\langle \langle a, b \rangle, c \rangle = -D(a)\langle b, c \rangle + (-1)^{|a||b|} D(b)\langle a, c \rangle.$$

Using the ideas of [26], we establish an analogue of Farkas' theorem. Namely, we prove that every PI unital generalized Poisson algebra and every PI unital algebra of Jordan brackets satisfy some customary identity (11).

For a homogeneous Lie word f with letters $x_1, x_2, \dots, x_n$ define the degree as the associative monomial $x_1^{\deg_{x_1}(f)} \dots x_n^{\deg_{x_n}(f)}$. Put in addition

$$\{x_1, x_2, \dots, x_n\} := \{\dots \{x_1, x_2\}, \dots, x_n\}.$$

Following [22], we easily obtain the following statement.

Lemma 8: *Given the free Lie algebra L with generators $1, y, x_1, \dots, x_n$, there is a basis for the component of degree $1^k y x_1 \dots x_n$ consisting of the elements*

$$\{y, y_{\sigma(1)}, \dots, y_{\sigma(n+k)}\},$$

where $y_i = x_i$ for $0 < i < n + 1$ and $y_i = 1$ for $n < i < n + k + 1$.

Lemma 9: In the above notation,

$$\{yz, w_1, \dots, w_n\} \\ = \sum \pm \{y, w_{i_1(1)}, \dots, w_{i_1(k_1)}\} \{z, w_{i_2(1)}, \dots, w_{i_2(k_2)}\} \{1, w_{i_3(1)}, \dots, w_{i_3(k_3)}\} \dots \{1, w_{i_l(1)}, \dots, w_{i_l(k_l)}\},$$

summed over all possibilities for which each subscript appears exactly once in each summand, all sequences $\{i_1(1), \dots, i_1(k_1)\}, \dots, \{i_l(1), \dots, i_l(k_l)\}$ are monotonely increasing, all elements $i_j(t)$ are different, and $1 \leq i_j(t) \leq n$, while $k_1 + \dots + k_l = n$. (In the case $p = 0$ the singleton $\{x\}$ is identified with x .)

Proof: An easy induction. □

Lemma 10: With the above notation,

$$\{w_1, \dots, w_r, y \cdot z, w_{r+1}, \dots, w_{r+n}\} \\ = \sum (\pm \{w_0, y, w_{i_1(1)}, \dots, w_{i_1(k_1)}\} \{z, w_{i_2(1)}, \dots, w_{i_2(k_2)}\} \cdot \{1, w_{i_3(1)}, \dots, w_{i_3(k_3)}\} \\ \dots \{1, w_{i_l(1)}, \dots, w_{i_l(k_l)}\} \\ \pm \{y, w_{i_1(1)}, \dots, w_{i_1(k_1)}\} \{w_0, z, w_{i_2(1)}, \dots, w_{i_2(k_2)}\} \cdot \{1, w_{i_3(1)}, \dots, w_{i_3(k_3)}\} \\ \dots \{1, w_{i_l(1)}, \dots, w_{i_l(k_l)}\} \\ \pm \{y, w_{i_1(1)}, \dots, w_{i_1(k_1)}\} \{z, w_{i_2(1)}, \dots, w_{i_2(k_2)}\} \cdot \{D(w_0), w_{i_3(1)}, \dots, w_{i_3(k_3)}\} \\ \dots \{1, w_{i_l(1)}, \dots, w_{i_l(k_l)}\}),$$

where $w_0 = \{w_1, \dots, w_r\}$.

Proof: For $r = 0$ this lemma follows immediately from the previous one. Hence, we assume that $r \geq 1$ and for $\beta = \{yz, w_0, w_{r+1}, \dots, w_{r+n}\}$ formally apply the previous lemma to this new expression. A simple calculation completes the proof. □

Given a polynomial $f(x, \dots)$, put

$$\Delta_{yz}^x(f) := f(yz, \dots) - yf(z, \dots) - zf(y, \dots).$$

It is easy to see that if $\Delta_{yz}^x(f) \neq 0$ and an algebra A satisfies the identity f then A also satisfies the identity $\Delta_{yz}^x(f)$.

Lemma 11: If f is a multi-linear Lie polynomial in the letters $x, w_1, \dots, w_m$ which is an associative algebraic derivation with respect to x when considered as a Poisson polynomial then $\deg(f) = 2$.

Proof: Suppose that f is of degree $1^k x w_1 \dots w_m$. Put $w_i = 1$ for $m < i < m + k + 1$ and $x = w_{m+k+1}$. By Lemma 8, we can write

$$h = \sum c_\sigma \{w_1, w_{\sigma(2)}, \dots, w_{\sigma(m+k+1)}\},$$

Where σ runs over the permutations of $2, \dots, m + k + 1$ and $c_\sigma \in F$. Calculate the derivation difference $\Delta_{yz}^{w_{m+k+1}}(h)$ with respect to $w_{m+k+1} \rightarrow yz$.

Some summands are the products of c_σ and

$$\{w_1, w_{\sigma(2)}, \dots, w_{\sigma(t-1)}, y\} \{z, w_{\sigma(t)}, \dots, w_{\sigma(m+k)}\}.$$

Other summands are the products of c_σ and $yzD\{w_1, w_{\sigma(2)}, \dots, w_{\sigma(m+k)}\}$. However, these expressions determine σ ; they do not appear in the derivation difference for any other permutation. Ordering the variables

$$y < z < w_1 < \dots < w_n,$$

we see that these special expressions are linearly independent for $\deg(f) > 2$ by Lemma 8. $\square$

Given a Poisson polynomial f , define x -height(f) as the maximal length of a $\{, \}$ -subword containing x .

Lemma 12: *If a multi-linear Poisson polynomial f is an associative derivation with respect to some letter x then x -height(f) < 3 .*

Proof: Write $f = \sum_\gamma f_\gamma \gamma$, where f_γ are Lie polynomials of degree $1^k x t$ for some t . Consider $f_\gamma = \sum_k f_\gamma^k$, where the degree of the polynomial f_γ^k is $1^k x t$. For f_γ^k choose polynomials with maximal $\deg(t)$ and, out of these, choose a polynomial f^* with maximal k . Suppose that its degree is $1^k x x_1 \dots x_n$. Continuing the argument as in Lemma 11, we infer that if $\deg(f^*) > 2$ then $\Delta_{yz}^x(f) \neq 0$. Then $\deg(f_\gamma) < 3$ and x -height(f) < 3 . $\square$

The main result of this chapter is the following theorem.

Theorem 13: *If a unital generalized Poisson algebra A satisfies a polynomial identity g_0 then A satisfies a polynomial identity g_* of type (11).*

Proof: Use the ideas of [26]. The proof goes in three steps.

Step 1 Assume that x -height(g_0) > 2 . Then A satisfies the identity $g_1 = \Delta_{yz}^x(g_0)$. Lemma 12 yields $g_1 \neq 0$. Observe that x -height(g_0) $> y$ -height(g_1) and x -height(g_0) $> z$ -height(g_1). If the polynomial g_1 has a letter w with w -height(g_1) > 2 then continue the process above. Eventually we see that A satisfies an identity f with v -height(f) < 3 for every letter v .

Step 2 Assume that A satisfies an identity f with x -height(f) < 3 for every letter x in f . Assume also that f is not an associative derivation with respect to any letter x . Then

$$f = xT + D(x)T_0 + \sum \{x, x_i\}T_i.$$

Consider

$$\Delta_{yz}^x(f) = -yzT + yz \sum D(x_i)T_i \neq 0.$$

It is easy to see that A satisfies the identity $f^* = -T + \sum D(x_i)T_i$ with fewer letters than f . Hence, if f^* is not a derivation with respect to any letter t then we can continue this process. It is finite, and we have shown that A satisfies an identity g with x -height(g) < 3 which is an associative derivation with respect to every letter.

Step 3 For a polynomial g with letters $x_1, \dots, x_n$, we can represent every associative subword of type $\{x, y\}$ as $\langle x, y \rangle + (D(x)y - xD(y))$. It follows that we can represent g as the polynomial $g_* + D(x_1)T_1 + x_1R_1$, where g_* is of type (11). It is easy to see that

g_* is an associative derivation with respect to every letter. Also it is easy to see that A satisfies the identity R_1 . It follows that A satisfies the identity $g_* + D(x_1)T_1$. Suppose that $D(x_1)T_1 = D(x_2)T_2 + x_2R_2$. Therefore, A satisfies the identity $g_* + D(x_1)D(x_2)T_{1,2}$. Continuing this process, we conclude that A satisfies an identity g_* of the form (11). $\square$

Consequently, we obtain the following statement.

Corollary 14: *Every PI generalized Poisson algebra satisfies an identity of the type*

$$f_* = \sum_{i=0}^{[m/2]} \sum_{\sigma \in S_m} c_{\sigma,i} \prod_{k=1}^i [x_{\sigma(2k-1)}; x_{\sigma(2k)}; z_{2k-1}; z_{2k}] \cdot \prod_{k=1}^{m-2i} \{x_{\sigma(2i+k)}; z_{2i+2k-1}; z_{2i+2k}\} \cdot \prod_{k=1}^{2i} z_{2m-2i+k}, \quad (12)$$

where

$$\begin{aligned} [u_1; u_2; w_1; w_2] &= \{u_1, u_2\}w_1w_2 + \{u_1, w_1w_2\}u_2 + u_1\{w_1w_2, u_2\} \\ &\quad + \sum_{\sigma_1, \sigma_2 \in S_2} \{u_{\sigma_1(1)}, w_{\sigma_2(1)}\}u_{\sigma_1(2)}w_{\sigma_2(2)}, \\ \{t_1; t_2; t_3\} &= \{t_2t_3, t_1\} - \{t_2, t_1\}t_3 - \{t_3, t_1\}t_2. \end{aligned}$$

Proof: By theorem 13, every PI generalized Poisson algebra satisfies an identity g_* of type (11). Multiplying g_* by $\prod_{k=1}^{2m} z_k$, observe that

$$w_1w_2\langle u_1, u_2 \rangle = [u_1; u_2; w_1; w_2] \text{ and } t_2t_3D(t_1) = \{t_1; t_2; t_3\},$$

establishing the corollary. $\square$

Also, we can prove an analogue of Farkas' theorem for PI algebras of Jordan brackets:

Theorem 15: *If a unital algebra of Jordan brackets A satisfies a polynomial identity g_0 then A satisfies polynomial identities of types (11) and (12).*

Proof: Consider the generalized Poisson algebra A_D on the vector space A with multiplications $\cdot$ and $\{\cdot, \cdot\}_D$. It is easy to see that A_D is a PI algebra. Using Theorem 13, we infer that A_D satisfies a polynomial identity g of type (11). Observing that $2D_D = D$ and

$$\{x, y\}_D + xD_D(y) - D_D(x)y = \{x, y\} + xD(y) - D(x)y,$$

we conclude that A satisfies a polynomial identity of type (11). Moreover, using the proof of Corollary 14, we see that A satisfies a polynomial identity of type (12). $\square$

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Appendix 1. The Kantor double for generic Poisson superalgebras

Let us give the definition of a generic Poisson superalgebra [9]. Consider a vector space GP equipped with the associative supercommutative multiplication $x \cdot y$ and the superanticommutative multiplication $\{x, y\}$. If the superalgebra $(GP, \cdot, \{, \})$ satisfies the Leibniz identity

$$\{x, y \cdot z\} = \{x, y\} \cdot z + (-1)^{|x||y|} y \cdot \{x, z\} \quad (A1)$$

then GP is a generic Poisson superalgebra (GP superalgebra). It is easy to see that if $(GP, \{, \})$ is a Lie superalgebra then $(GP, \cdot, \{, \})$ is a Poisson superalgebra. It is known that the Kantor double of a Poisson superalgebra is a Jordan superalgebra [11]. The following question arises naturally.

Question: When is the Kantor double of a generic Poisson superalgebra a Jordan superalgebra?

We answer this question in the following assertion.

Theorem 16: *Given a GP superalgebra A over a field $\mathbb{F}$ of characteristic $\neq 2, 3$, the Kantor double $K(A)$ is a Jordan superalgebra if and only if A satisfies the special identity*

$$(\{ \{a, b\}, c\} - (-1)^{|b||c|} \{ \{a, c\}, b\} - \{a, \{b, c\}\})d = 0 \quad (A2)$$

for all elements $a, b, c, d \in GP_0 \cup GP_1$.

Proof: In the case that $(B, \cdot)$ is an associative supercommutative superalgebra and $(B, \{, \})$ is a superanticommutative superalgebra, [15] shows that $K(B)$ is a Jordan superalgebra if and only if the homogeneous elements of the superalgebra B satisfy the identities

$$\begin{aligned} & (-1)^{(k+j)i} (\{h_k k_l, g_j\} f_i - h_k k_l \{g_j, f_i\}) \\ &= (-1)^{(l+j)k} (\{k_l f_i, g_j\} h_k - k_l f_i \{g_j, h_k\}), \\ & (-1)^{(i+j)l} (\{f_i h_k g_j, k_l\} - f_i h_k \{g_j, k_l\}) \\ &= (-1)^{(k+j)i} (\{h_k k_l, g_j\} f_i - \{h_k k_l, g_j f_i\}) \end{aligned} \quad (A3)$$

$$\begin{aligned} &+ (-1)^{(l+j)k} (\{k_l f_i, g_j\} h_k - \{k_l f_i, g_j h_k\}), \\ & (-1)^{(i+j)l} \{f_i, h_k\} g_j k_l + (-1)^{(k+j)i} \{h_k, k_l\} g_j f_i + (-1)^{(l+j)k} \{k_l, f_i\} g_j h_k \\ &= (-1)^{(i+j)l} \{f_i, h_k\} \{g_j, k_l\} \end{aligned} \quad (A4)$$

$$+ (-1)^{(k+j)i} \{h_k, k_l\} \{g_j, f_i\} + (-1)^{(l+j)k} \{k_l, f_i\} \{g_j, h_k\}, \quad (A5)$$

where $k_i, h_i, g_i, f_i \in B_i$.

It is easy to see that if an algebra satisfies (A1) then it satisfies (A3) and (A4). Combining (A1) and (A5), we obtain (A2). $\square$

Corollary 17: *If A is a unital GP (non-Poisson) superalgebra then $K(A)$ is not a Jordan superalgebra.*

Corollary 18: *If A is a GP (non-Poisson) superalgebra and $(A, \cdot)$ is an algebra with trivial annihilator (for example, a prime algebra) then $K(A)$ is not a Jordan superalgebra.*

Example 19: Consider an anticommutative algebra A with multiplication $\{, \}$ which is not a Lie algebra (for instance, A Malcev algebra or a binary Lie algebra). Define the new multiplication as $x \cdot y = 0$. Then $(A, \cdot, \{, \})$ is a GP algebra which is not a Poisson algebra and satisfies (A2), and $K(A)$ is a Jordan superalgebra.